KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2015
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

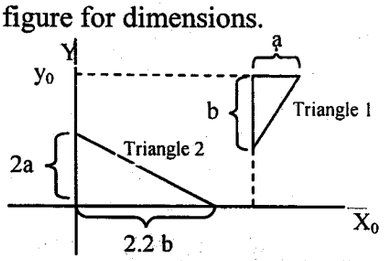
TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What are the raster and vector graphics? Write the differences between vector and raster graphics. (07)
   b) We wish to scale a triangle A(0,0), B(1,1), C(5,2) to twice its size. While keeping vertex C fixed. This scaling can be expressed as a composite transformation. Describe and write the matrices for the component transformations. (10)
   c) Transform Triangle1 to Triangle2 in the following figure by a sequence of basic transformations. Write the transformation matrices in 2D Homogeneous coordinates. Compute the final transformation matrix. Note that the size of triangle as also been changed; see the figure for dimensions. (13)

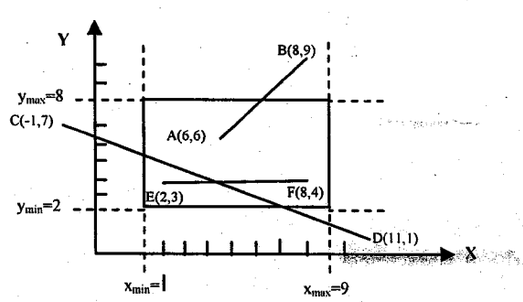


   d) What are the drawbacks of DDA algorithm over Bresenham's line drawing algorithm. (05)

2. a) Scan convert the line segment from point (5,8) to point (9,12). (10)
   b) Why is the point to point line error always $\leq 1/2$ for the mid-point scan conversion algorithm? (10)
   c) Prove that $R(\theta_1).R(\theta_2)=R(\theta_1+\theta_2)$ Where R( ) is a rotation function and θ_1, θ_2 are angle of rotation. (08)
   d) Convert these homogeneous points to Cartesian (0,1,3,2),(2,1,3,4),(2,4,3,5). (07)

3. a) When performing Cohen-Sutherland line clipping, how do we use outcodes to check for trivial rejection (what exact operation and comparison would you use?). (06)
   b) What are the new coordinates of the diamond shaped polygon whose vertices are A(-1,0), B(0,-2), C(1,0) and D(0,2). About the horizontal line y=2 after the reflection. (11)
   c) Derive the perspective projection matrix. In a perspective projection matrices with z_0 =6, scene point (-20,8,15) projects to what (x,y) point on the virtual image plane? To what (x,y) point on the real image plane? (11)
   d) Describe the traditional graphics pipeline and explain what is done in each stage. (07)

4. a) Use the Liang-Bersky algorithm to clip the lines in following figure (12)



b) What are the steps to calculate a transformation matrix to find the mirror reflection of an object with respect to an arbitrary plane whose normal is N and a point P is on the plane. (10)

c) Derive a 3D transformation matrix to rotate an object by θ degree with respect to z-axis. (09)

d) State the name of different types of parallel and perspective projections. (04)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Why do we need illumination models? Derive the Phong specular reflection model and combine diffuse and specular reflection. (13)

b) What is error diffusion? Write the algorithm for error diffusion image representation. (08)

c) Describe the strength and weakness of Ray tracing, using the theory of "The Rendering Equation". (06)

d) Compare between Phong surface rendering method and Gouraud surface rendering method. Which is better and why? (08)

6. a) Consider a triangle which has three vertices as $P_1(1,2)$, $P_2(5,0)$ and $P_3(3,4)$. A scan line S go through it and intersection points are $P_4(3,1)$ and $P_5(4,1)$. Inside the triangle there is a point P(3.5,1) on scan line. Normals for three vertices are $N_1(0,1,0)$,$N_2(0,0,1)$ and $N_3(0,0.7,0.7)$. We also have the following values: L=(0,0.7,0.7), v=(0,0,1) ,K_a=0.1,I_a=0.5,K_d=0.5, K_s=0.2, n_s=1,I_L=0.6, where symbols denote usual meanings. Find Intensity I_4, I_5, and I_p on scan line with Gouraud shading. (22)

b) Prove that reflected ray path, (13)

T=$\frac{n_i}{n_r}u - (\cos\theta_r - \frac{n_i}{n_r}Cos\theta_i)$n, where u is incoming ray and n is normal.

7. a) Let S_1, be a sphere of radius 8 centered at (2,4,1) and S_2 a sphere of radius 10 centered at (10,-2,-5). Determine if a ray with s=2J+5K and d=I-2K intersects the spheres and draw necessary diagram to show the intersections. (14)

b) What do you mean by depth buffer method and z-fighting? (06)

c) Define Hermite Surface. Find geometry vector G_{HX} for Hermite surface. Also find twist of the vector. (10)

d) What is terrain in computer graphics? How do you generate terrain? (05)

8. a) What is particle system? How particles are generated? (07)

b) Define fractal. What are the classifications of fractal objects? How is fractal generated? (09)

c) What do you know about geometric continuity? Explain G^0, G^1 and G^2 geometric continuity? (08)

d) Explain briefly about natural cubic splines and cubic Bezier curves. (11)